Multiplying powers of the same number

To multiply powers that have the same base number, simply add the powers. The power of the answer is the sum of the powers that are being multiplied.

the first power · add the powers · the second power

$$6^2 \times 6^3 = 6^5$$

the power of the answer is: 2 + 3 = 5

because

▷ **Writing it out**
Writing out what each of these powers represents shows why powers are added together to multiply them.

$$(6\times6)\times(6\times6\times6) = 6\times6\times6\times6\times6$$

6^2 is 6×6 · 6^3 is $6 \times 6 \times 6$ · $6 \times 6 \times 6 \times 6 \times 6$ is 6^5

Dividing powers of the same number

To divide powers of the same base number, subtract the second power from the first. The power of the answer is the difference between the first and second powers.

the first power · subtract the second power from the first · the second power

$$4^4 \div 4^2 = 4^2$$

the power of the answer is: 4 – 2 = 2

because

▷ **Writing it out**
Writing out the division of the powers as a fraction and then canceling the fraction shows why powers to be divided can simply be subtracted.

4^4 is $4 \times 4 \times 4 \times 4$

$$\frac{4\times4\times4\times4}{4\times4} \rightarrow \frac{\cancel{4}\times\cancel{4}\times4\times4}{\cancel{4}\times\cancel{4}} = 4\times4$$

4^2 is 4×4 · cancel the fraction to its simplest terms · 4×4 is 4^2

LOOKING CLOSER

Zero power

Any number raised to the power 0 is equal to 1. Dividing two equal powers of the same base number gives a power of 0, and therefore the answer 1. These rules only apply when dealing with powers of the same base number.

the first power · the second power · the power of the answer is: 3 – 3 = 0

$$8^3 \div 8^3 = 8^0 = 1$$

any number to the power 0 = 1

because

▷ **Writing it out**
Writing out the division of two equal powers makes it clear why any number to the power 0 is always equal to 1.

$$\frac{8\times8\times8}{8\times8\times8} = \frac{512}{512} = 1$$

8^3 is $8 \times 8 \times 8$ · any number divided by itself = 1